Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer pursuing an MSCS at USC with 2+ years building **scientific data workflows**, **Python/Java services**, and AI-enabled applications, with experience across pipelines, APIs, cloud deployments, and cross-functional technical communication.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed and optimized multi-tenant scientific data workflows supporting 500+ users, reducing latency from **1.2s to 350ms** through SQL tuning, indexed access patterns, and backend service improvements.
- Built ETL-style integration pipelines processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases, improving data quality, reliability, and diagnosis of runtime anomalies.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed full-stack React and Flask workflows handling 10,000+ daily requests, translating user feedback into maintainable APIs, responsive interfaces, and iterative enhancements validated directly in production.
- Built AI-assisted grading workflows with multimodal models, OCR, and relational persistence, reducing feedback turnaround by **60%** while supporting structured data capture and operational handoffs.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built Python experimentation pipelines across 100,000+ trajectory records, enabling reproducible evaluation, simulation analysis, and benchmark consistency for model-driven planning and optimization workloads.
- Produced technical reports and validation scenarios translating research findings into actionable improvements, strengthening cross-functional alignment, reducing ambiguity, and supporting clearer engineering decisions with scientists and engineers.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable data movement and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence during collaborative release cycles.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built a MERN-based operations platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular services and responsive user-facing interfaces.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing friction during validation and troubleshooting.

Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, C++, Bash
Data & AI: ETL pipelines, data validation, LLM applications, RAG concepts, agentic workflow familiarity, graph/NoSQL concepts
Backend & APIs: FastAPI, Flask, REST APIs, microservices, event-driven concepts, PostgreSQL, SQL Server, MongoDB
Cloud & Delivery: AWS, Docker, Kubernetes, Git/GitHub, GitHub Actions, Jenkins, CI/CD, Argo familiarity, cross-functional collaboration

Projects

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*

- Built a secure document platform with semantic search, summarization, and collaborative Q&A, demonstrating RAG-style workflows over 50K+ documents while reducing document-analysis time by **60%**.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an AI-powered SQL workspace supporting 10+ databases, maintaining Java and TypeScript modules that accelerated schema exploration, query authoring, and structured debugging of data workflows.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a containerized RESTful data processing API with automated CI/CD releases and validation checks, improving deployment reliability, interoperability, and supportability for production data workflows.